Problem Solving & C Programming Fundamentals

Course: C Programming | Semester 1 | Contributor: Priya Sharma

From pointers and dynamic memory allocation to struct alignments and file I/O operations.

Core Concept 1: Pointers & Pointer Arithmetic

* Understanding l-values, r-values, and memory address operators & and *.
* Pointer arithmetic: incrementing ptr shifts by sizeof*ptr bytes.
* Array decay and passing 2D arrays to functions using pointer to arrays.

Core Concept 2: Dynamic Memory & Structure Padding

* malloc vs calloc vs realloc with proper free and dangling pointer avoidance.
* Structure memory layout, boundary alignment, and packing pragmas.
* Self-referential structures: foundational linked list node definitions.

Important Exam Gotchas

* Never return pointer to automatic local stack variable.
* Always check malloc return against NULL before dereferencing.

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